

GOKULA-HEALTH

Digital Health Card for Cattle — Dairy Livestock Management App

Android App Development Using Generative AI | Project Requirements Document

Domain: Agriculture / Livestock	Platform: Android (Kotlin / XML)	Storage: Room DB (SQLite)	Gen AI: Gemini API (Health Advisory)
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1. PROBLEM STATEMENT

India is the world's largest milk producer, yet small-scale dairy farmers continue to face significant income losses due to preventable management gaps. The two most critical — and most commonly missed — events in a dairy cow's life are the **Heat Cycle** (the narrow fertile window for breeding) and **vaccination schedules** (especially for Foot and Mouth Disease). Missing either results in reduced herd size, disease outbreaks, and direct economic loss.

Livestock health management in rural India is almost entirely oral and memory-based. There is no structured digital record. This means when a cow is sold or insured, there is no verifiable health history — reducing the animal's market value. **Gokula-Health** replaces the farmer's notebook with a smart, offline-capable digital health passport for every animal.

■ Missed Heat Cycles Breeding windows of 12–18 hours missed due to lack of cycle tracking, reducing annual calf births.	■ Skipped Vaccinations FMD outbreaks cost farmers ■20,000+ per animal due to missed scheduled doses.	■ Yield Decline Ignored Gradual milk drop goes unnoticed without daily logging and trend visualisation.	■ No Health Records No documented health history reduces resale / insurance value of cattle by 15–30%.
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2. SCOPE OF THE APP

In Scope

- Android app for small-to-medium dairy farmers to manage individual cattle health records.
- Cattle profile creation with photo, Ear Tag ID, breed, age, and ownership details.
- Daily milk yield diary — morning and evening entries (in litres) per animal.
- Automatic monthly average yield calculation per cattle.
- 30-day milk yield trend chart using MPAndroidChart library.
- Vaccination scheduler with automated AlarmManager-based reminders for FMD and other shots.
- Heat cycle calendar — tracking and alerting for upcoming fertile windows.
- Offline-first local storage using Room DB — no internet required for core features.
- AI-powered health advisory using Gemini API: suggest possible illness based on yield drop pattern.
- Health Passport export — printable/shareable summary of a cattle's full health record.
- Intuitive icon-driven UI: Cow ■, Milk ■, Syringe ■ icons for low-literacy farmer usability.

Out of Scope (v1.0)

- X Veterinary doctor booking or telemedicine features.

- ✗ Marketplace for cattle buying/selling.
- ✗ Government subsidy tracking or NABARD integration.
- ✗ iOS or web dashboard builds.
- ✗ Multi-farm / cooperative-level herd management (future roadmap).
- ✗ Feed and fodder cost tracking (planned for v2.0).

3. FUNCTIONAL REQUIREMENTS

ID	Feature	Description
FR-01	Cattle Registration	The app shall allow farmers to register each cattle with: Photo (CameraX), Ear Tag ID, Name, Breed, Date of Birth, and Gender. Each record stored in Room DB.
FR-02	Cattle Profile View	The app shall display a full profile card for each cattle showing all registered details, current health status, and quick-action buttons (Add Milk Entry, Add Vaccination, View Chart).
FR-03	Milk Yield Diary	The app shall allow twice-daily milk entry (Morning / Evening) per cattle in litres. Each entry timestamped and stored locally in Room DB.
FR-04	Monthly Average Yield	The app shall automatically calculate and display the Monthly Average Milk Yield per cattle by averaging all daily entries in the current calendar month. No manual calculation required.
FR-05	30-Day Yield Chart	The app shall render a line chart (MPAndroidChart) showing daily total milk yield (morning + evening) for the past 30 days per cattle. Chart must update in real time on new entry.
FR-06	Vaccination Scheduler	The app shall allow addition of vaccination events with: Vaccine Name, Date Administered, Next Due Date, and Notes. AlarmManager must trigger a notification on the due date regardless of internet connectivity.
FR-07	Heat Cycle Tracker	The app shall allow logging of observed heat cycle dates and calculate the projected next heat date (default cycle: 21 days). A reminder notification shall be sent 2 days before projected date.
FR-08	Offline-First Operation	All core features — cattle profiles, milk diary, vaccination records, reminders — must function completely without internet. Room DB is the single source of truth.
FR-09	AI Health Advisory	The app shall send the last 7 days of milk yield data to Gemini API and display a plain-language health observation (e.g., 'Yield has dropped 20% — consider veterinary check').
FR-10	Health Passport Export	The app shall generate a formatted summary (PDF or shareable text) of a cattle's complete health history: profile, vaccinations, yield trend, and breeding record.
FR-11	Herd Overview Dashboard	The home screen shall display a summary of all registered cattle: count, how many have pending vaccinations, and which cows are in or near heat cycle.
FR-12	Reminder Management	The app shall list all upcoming reminders (vaccination + heat cycle) in a unified calendar-style view. Users can dismiss, reschedule, or mark as done.

4. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement	Target / Metric
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Performance	App cold-start time	< 2 seconds on Android 8+ mid-range device
Performance	Chart render time (30-day)	< 1 second after data load
Performance	Milk entry save time	< 500ms, including Room DB write
Reliability	Offline alarm reliability	AlarmManager reminders fire 100% without internet
Reliability	Data persistence	No data loss on app kill, reboot, or OTA update
Reliability	AI advisory availability	Graceful fallback message shown if Gemini API unavailable
Usability	Icon-driven UI	Every primary action accessible via a recognisable icon (■■■■)
Usability	Low-literacy support	Core flow completable without reading text — icons + colour cues
Usability	Onboarding time	First cattle registered in under 3 minutes by a first-time user
Security	Local data	All Room DB data stored in app-sandboxed storage; no cloud upload of PII
Security	AI data	Only aggregated yield numbers (not farmer identity) sent to Gemini API
Scalability	Herd size	App performs without degradation for up to 50 cattle per farmer
Battery	Background alarms	AlarmManager used only for scheduled reminders; no continuous background service
Compatibility	Android support	Android 8.0 (API 26) and above; tested on low-RAM (2GB) devices
Language	Localisation	UI strings externalised to strings.xml; ready for Kannada / Hindi translation

5. FEATURE PRIORITISATION

5.1 Must-Have Features (MVP — Release Blockers)

■ Cattle Registration & Profile

Register each animal with photo, Ear Tag ID, breed, and DoB. View full profile card with all health details. Room DB-backed; works 100% offline.

■ Milk Yield Diary

Twice-daily (AM/PM) litre entries per animal. Auto-calculates daily total and monthly average. No internet required.

■ 30-Day Yield Chart

MPAndroidChart line graph showing milk yield trend over 30 days. Visual cue for yield drops requiring veterinary attention.

■ Vaccination Scheduler + Alarms

Add vaccination records with next-due-date. AlarmManager fires reminder notification on due date — offline guaranteed. FMD and other vaccines supported.

■ Heat Cycle Tracker

Log observed heat dates. App auto-calculates next expected heat (21-day cycle) and sends reminder 2 days prior via AlarmManager.

■ Herd Overview Dashboard

Home screen showing: total cattle count, pending vaccinations count, cattle near heat cycle — all in icon-driven card layout.

■ Fully Offline Operation

Every core feature (profiles, diary, charts, reminders) must work without internet. Room DB is sole data source.

5.2 Good-to-Have Features (v1.1+ Enhancements)

■ **Gemini AI Health Advisory** — Send last 7 days of yield data to Gemini API. Display plain-language advice: 'Yield dropped 25% — check for mastitis.' Helps farmers act before consulting a vet.

■ **Health Passport Export** — Generate a shareable PDF or WhatsApp-ready summary of the cattle's full health history — ideal for resale, insurance, or bank loan applications.

■ **Illness Symptom Logger** — Farmer can log observed symptoms (fever, limping, loss of appetite). Gemini API cross-references with yield trends to suggest likely condition.

■ **Cattle Photo Gallery** — Per-animal photo timeline showing physical growth or visible health changes over months.

■ **Cloud Backup (Optional)** — Firebase Firestore optional sync for backup across phone replacements. Farmer controls opt-in.

■ **Voice Input Support** — Voice-to-text milk entry for farmers uncomfortable with typing. Regional language support (Kannada, Hindi).

■ **Breeding History Log** — Track calving dates, bull/AI breeding records, and gestation countdown timer (283 days for cattle).

■ **Yield Valuation Calculator** — Auto-calculate estimated daily/monthly dairy income based on current yield and local milk rate (farmer-configurable ■/litre).

■ **Farmer Insights Report** — Monthly auto-generated report: best-performing cow, average herd yield, missed vaccinations — shareable with dairy cooperative.

■ **Smart Notification Grouping** — Group reminders by urgency: Critical (vaccination overdue) → Warning (heat cycle in 2 days) → Info (monthly report ready).

6. APP USER FLOW

Step	Screen	Action / Description
01	LAUNCH	Splash → Language selection (first run) → Permission: Camera + Notifications → Home Dashboard
02	HOME DASHBOARD	Cards: Total Cattle Pending Vaccinations Cows Near Heat Cycle Quick Add buttons

03	ADD CATTLE	Enter Ear Tag ID → Take photo (CameraX) → Fill breed, DoB, name → Save to Room DB
04	CATTLE PROFILE	View full details → Tabs: Milk Diary Vaccinations Heat Cycle AI Advisory
05	ADD MILK ENTRY	Select cattle → Choose Morning / Evening → Enter litres → Save → Chart auto-updates
06	VIEW YIELD CHART	MPAndroidChart 30-day line graph per cattle → Monthly average shown below chart
07	ADD VACCINATION	Select vaccine name → Date administered → Set next due date → AlarmManager schedules reminder
08	HEAT CYCLE LOG	Enter last observed heat date → App calculates next expected → Sets reminder 2 days before
09	AI ADVISORY	Tap 'Get Health Check' → App sends yield data to Gemini API → Displays advice card
10	HEALTH PASSPORT	Tap 'Export' on cattle profile → Generates shareable health summary (PDF/text)

7. TECHNICAL ARCHITECTURE & IMPLEMENTATION

Layer	Technology	Purpose
UI Layer	XML Layouts + ViewBinding / Jetpack Compose	Farmer-friendly, icon-driven UI; RecyclerView for cattle list
Camera	CameraX (androidx.camera)	Cattle photo capture; lifecycle-aware on all Android 8+ devices
Charts	MPAndroidChart	30-day milk yield line chart per cattle; real-time updates
Local DB	Room DB (SQLite)	All cattle profiles, milk entries, vaccination records; offline-first
Scheduling	AlarmManager + BroadcastReceiver	Vaccination and heat cycle reminders; fire without internet
Architecture	MVVM + Repository Pattern	ViewModel, LiveData/StateFlow, UseCases; Hilt Dependency Injection
Gen AI	Gemini 1.5 Flash API (Text)	Health advisory from yield trend data; graceful offline fallback
Photo Storage	Internal App Storage	Cattle photos stored privately in app files directory
Export	iText / Android PDF API	Health Passport generation as shareable PDF
Navigation	Jetpack Navigation Component	Fragment-based navigation with safe args
Background	WorkManager (optional sync)	Optional Firebase backup if farmer opts into cloud sync

8. IMPACT GOALS

■ White Revolution 2.0

India produces 220+ million tonnes of milk annually, yet per-animal productivity lags global averages. Data-driven management through Gokula-Health can increase per-cow yield by an estimated 10–15% through timely interventions and early illness detection.

■ **Farmer Income Protection**

A single missed FMD vaccination can wipe out an entire herd and cost a small farmer ■2–5 lakh. Ensuring 100% vaccination coverage through automated reminders directly protects household income.

■ **Livestock Asset Value**

A cattle with a verifiable digital health passport commands 15–30% higher resale value and is more easily insurable under government schemes (PM Fasal Bima equivalent for livestock). Gokula-Health creates this record automatically.

■ **Rural Digitisation**

Gokula-Health brings precision agriculture technology to the last mile — the individual farmer with 2–5 cows. It does not require smartphone expertise, internet access, or literacy to operate the core features.

■ **Policy Data Foundation**

Aggregated (anonymised) herd health data from many farmers can feed into district-level animal husbandry dashboards, helping government agencies track disease outbreaks and vaccination coverage in real time.

9. SUCCESS CRITERIA FOR STUDENTS

ID	Criterion	Priority	Acceptance Condition
SC-01	Auto Monthly Average Yield	Critical	The app must automatically compute and display Monthly Average Yield per cattle from all milk entries in the current month — no manual summation.
SC-02	Offline Reminders	Critical	Vaccination and heat cycle reminders via AlarmManager must fire correctly even when the device has no internet connectivity.
SC-03	Icon-Driven UI	Critical	All primary actions must use recognisable icons (■ Cattle, ■ Milk, ■ Syringe). The core reporting flow must be completable without reading any text.
SC-04	30-Day Yield Chart	Critical	MPAndroidChart must render a line graph of daily milk yield for each cattle for the past 30 days. Chart must update immediately after a new entry is saved.
SC-05	Cattle Registration	Critical	Each cattle must be registerable with photo, Ear Tag ID, breed, and DoB. Profile must be retrievable after app restart (Room DB persistence verified).
SC-06	Heat Cycle Calculator	High	App must correctly calculate next expected heat date (last observed date + 21 days) and trigger a reminder 2 days before projected date.
SC-07	Vaccination Record	High	Adding a vaccination must store the record and schedule an AlarmManager notification for the next due date. Notification must show cattle name and vaccine name.
SC-08	AI Health Advisory	Medium	Gemini API integration must return a readable health observation based on 7-day yield data. A fallback message must display if API is unavailable.

SC-09	Health Passport Export	Medium	A shareable health summary must be generated for any registered cattle containing profile, vaccination history, and yield trend.
SC-10	Code Architecture	Medium	MVVM pattern enforced: ViewModel must not hold Context. Repository layer separates Room DB access from UI. Hilt used for dependency injection.